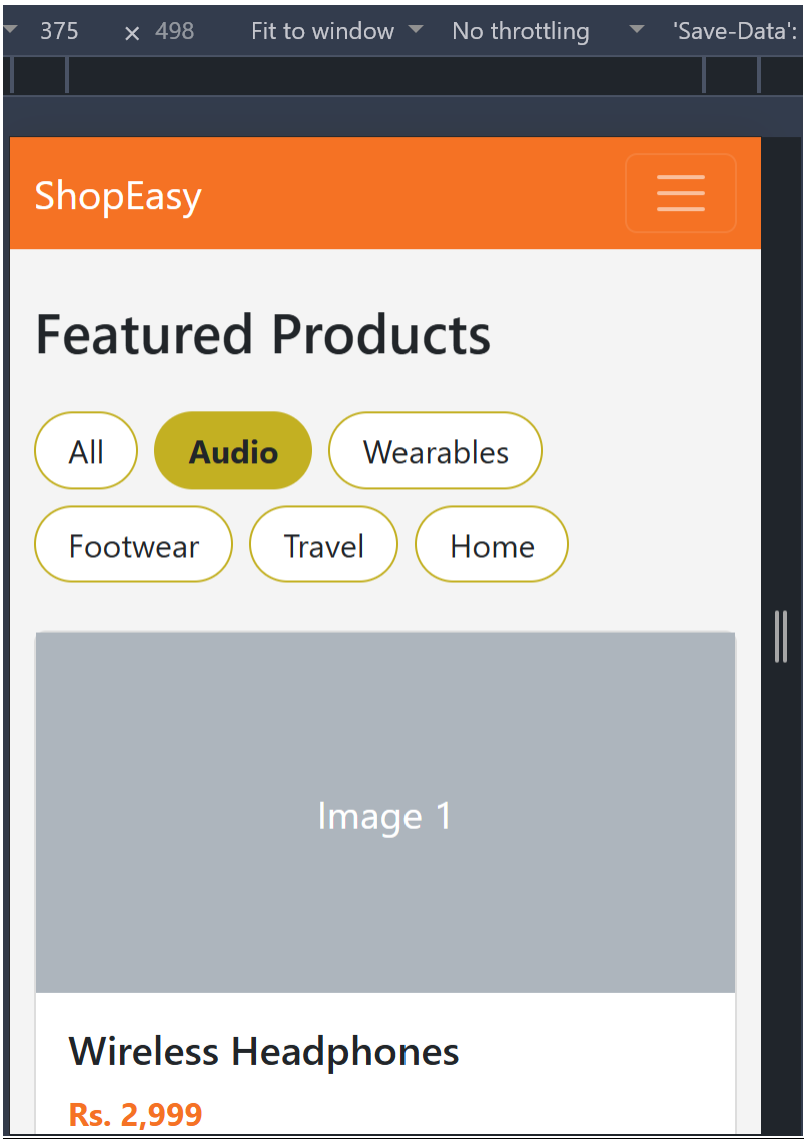
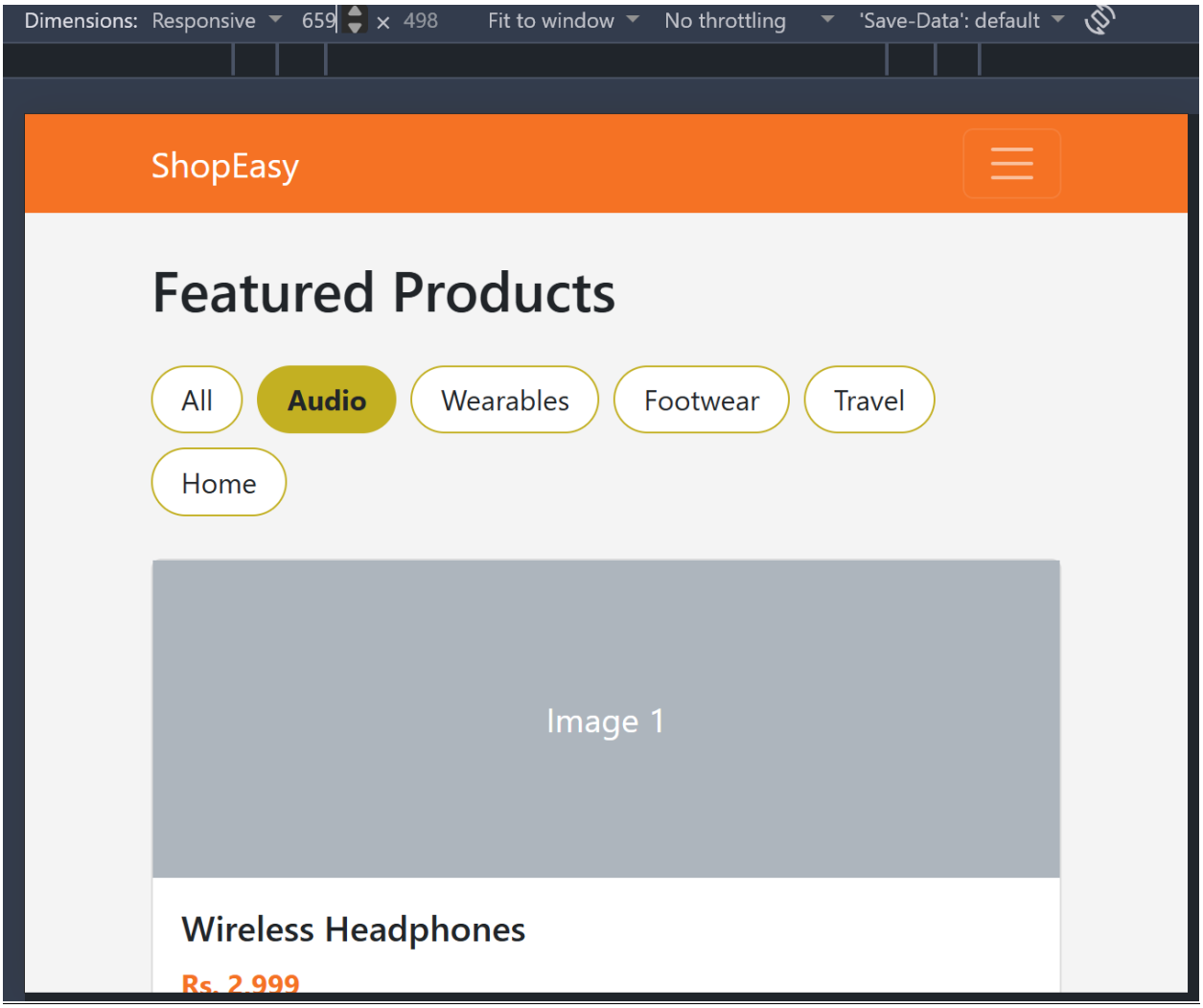
Name: Abdullah Reg. No: FA24-BCS-159

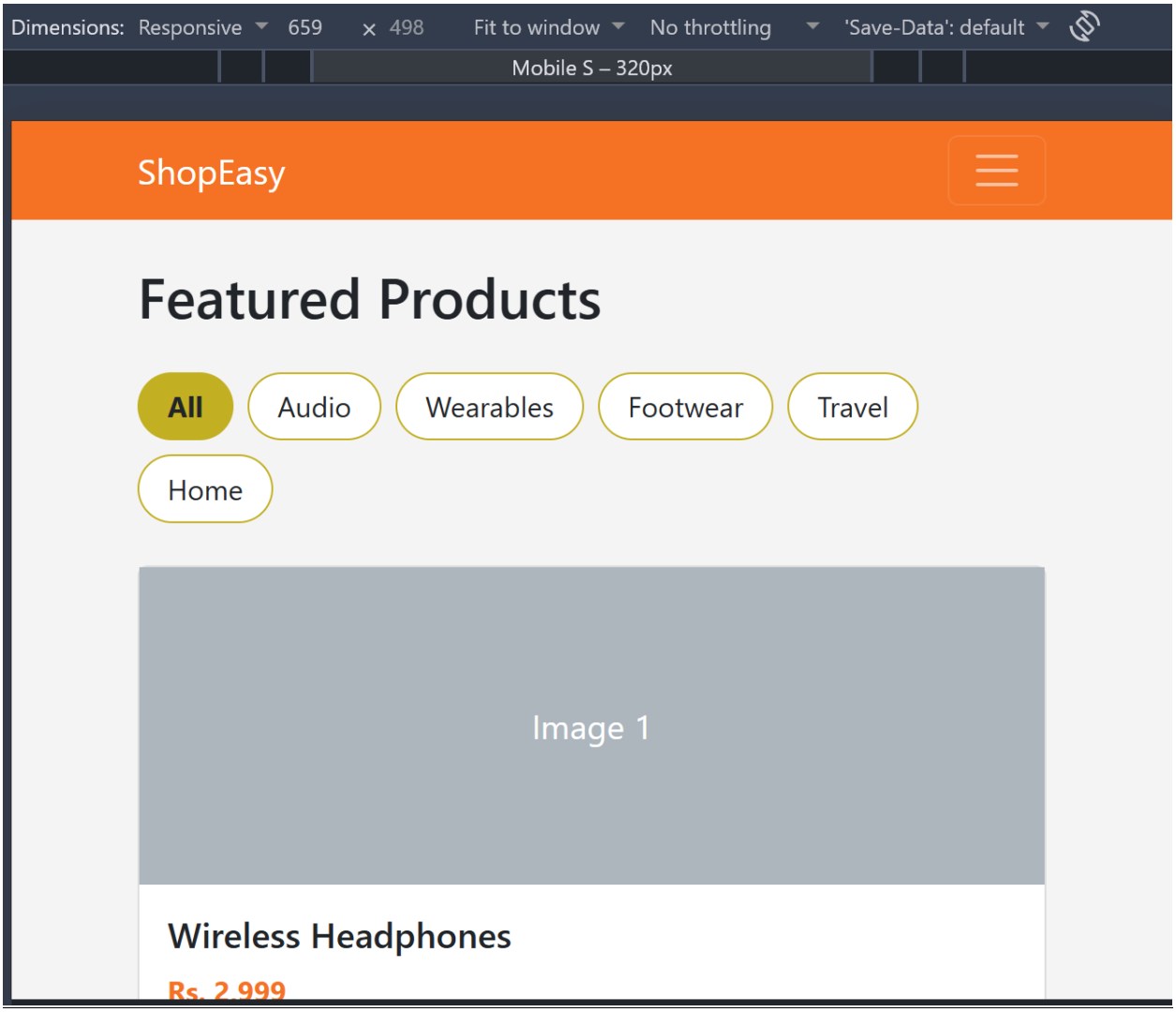
Web tech

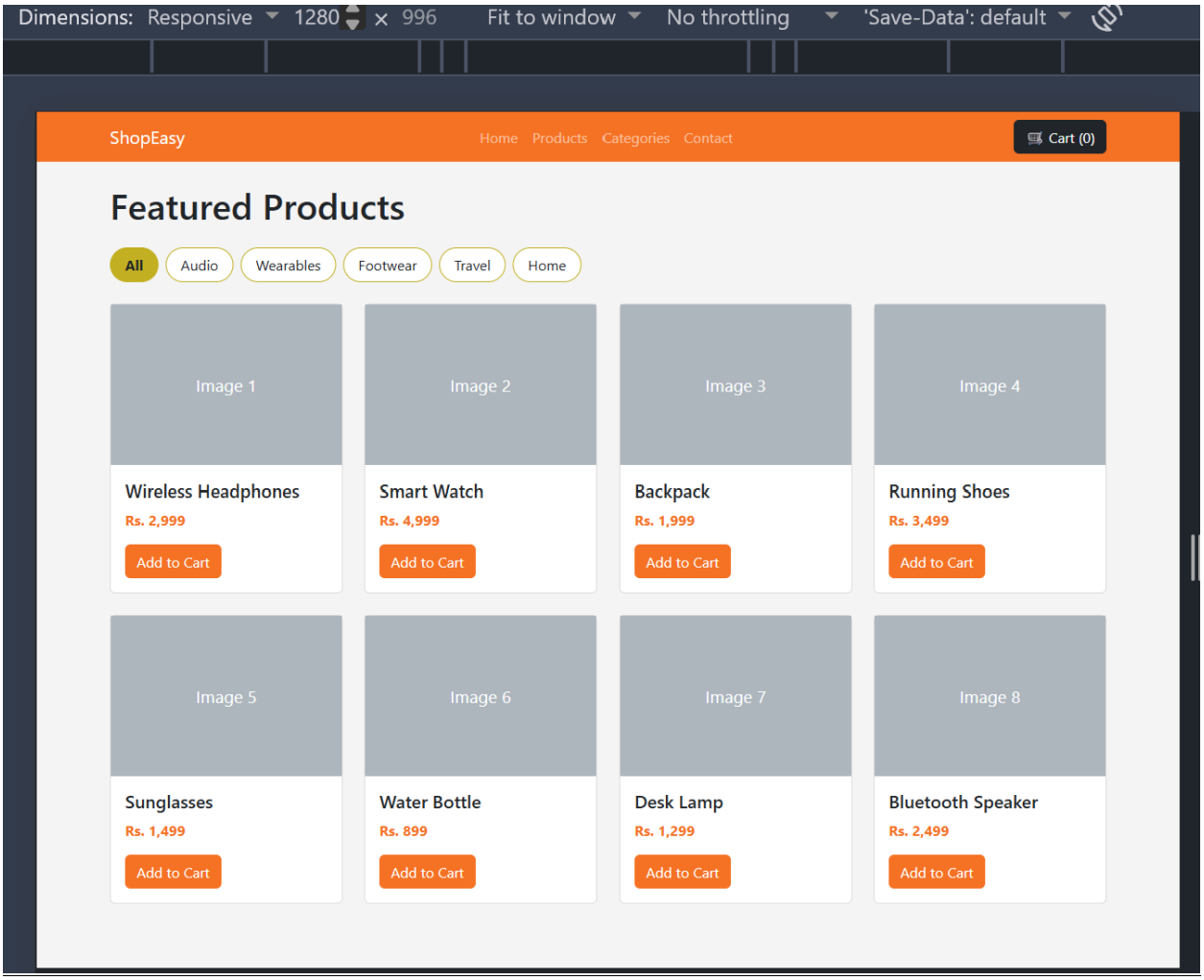
Completion Screen shots:

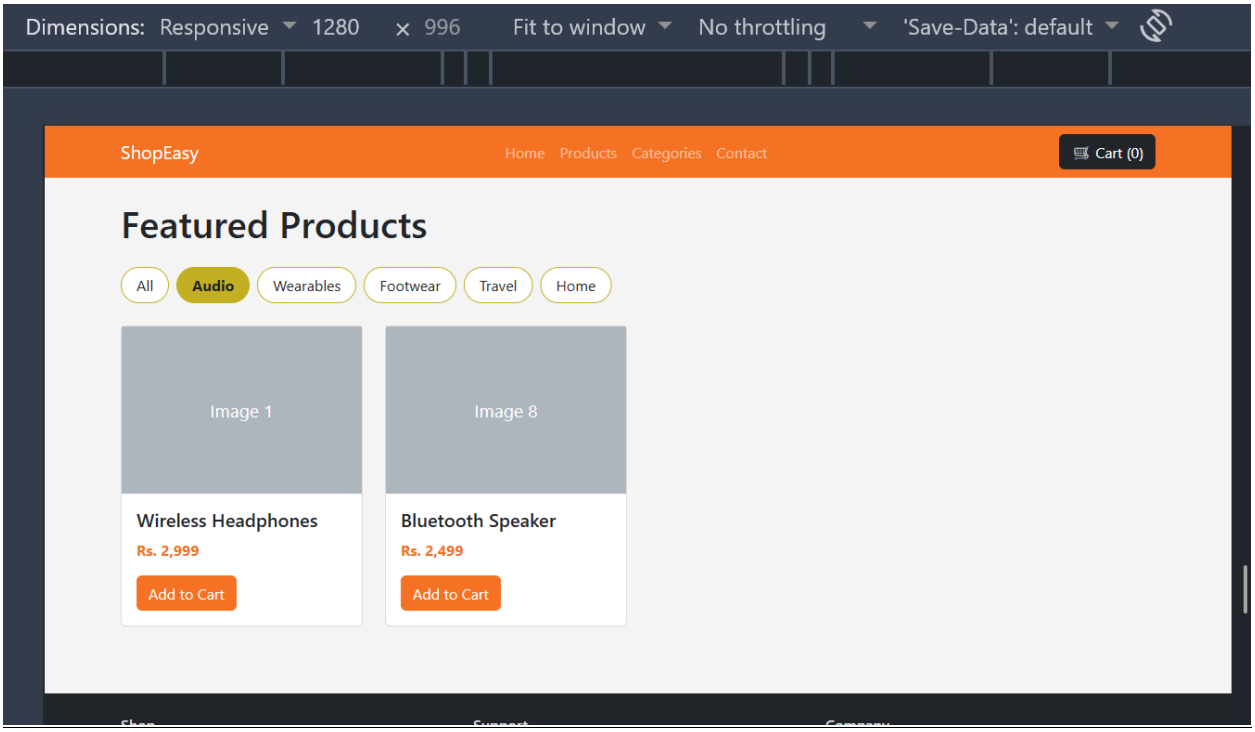
Task 1:



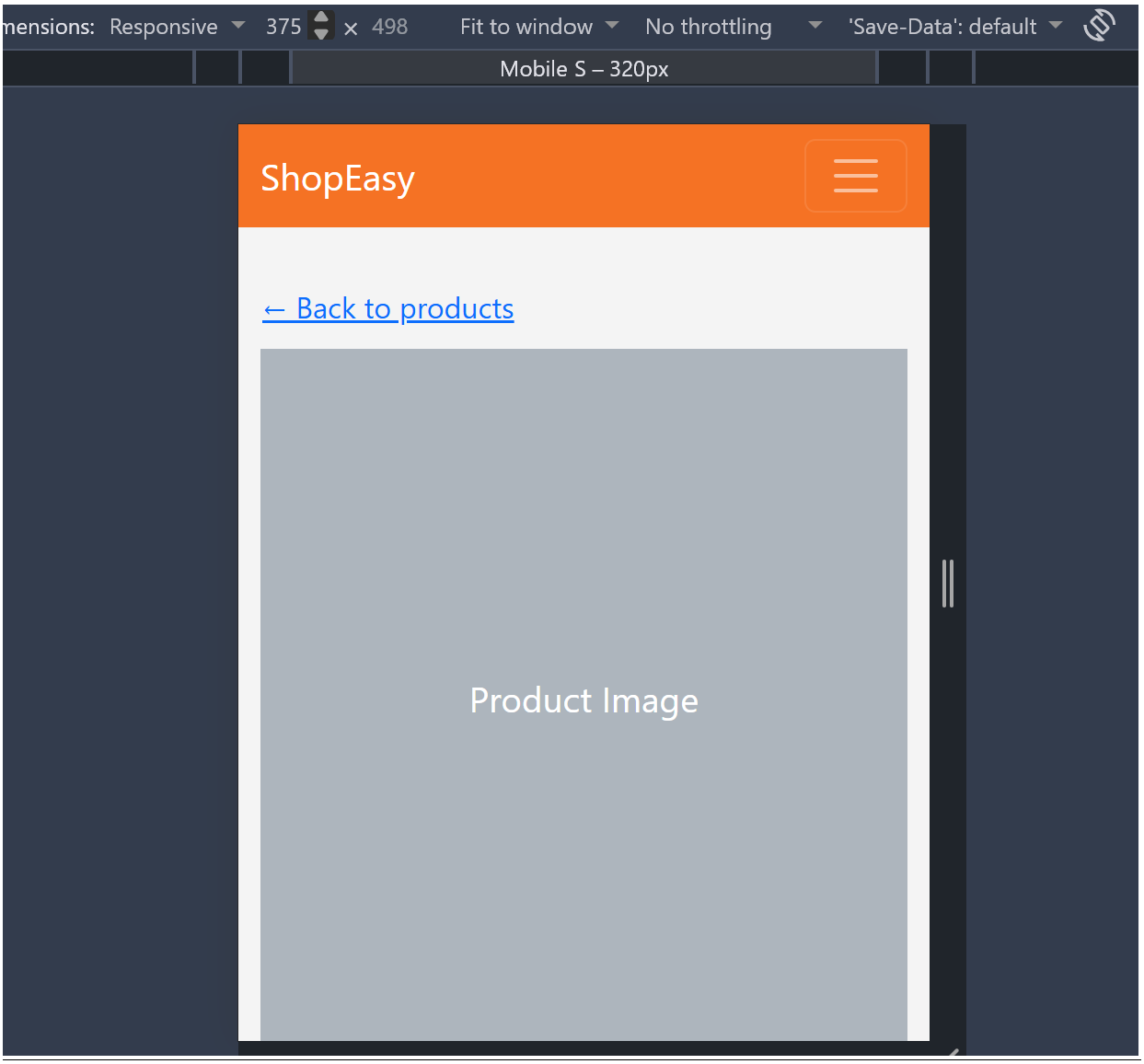

Dimensions: Responsive 659 × 498 Fit to window No throttling 'Save-Data': default
ShopEasy
Featured Products
All
Audio
Wearables
Footwear
Travel
Home
Image 1
Wireless Headphones
Rs. 2,999

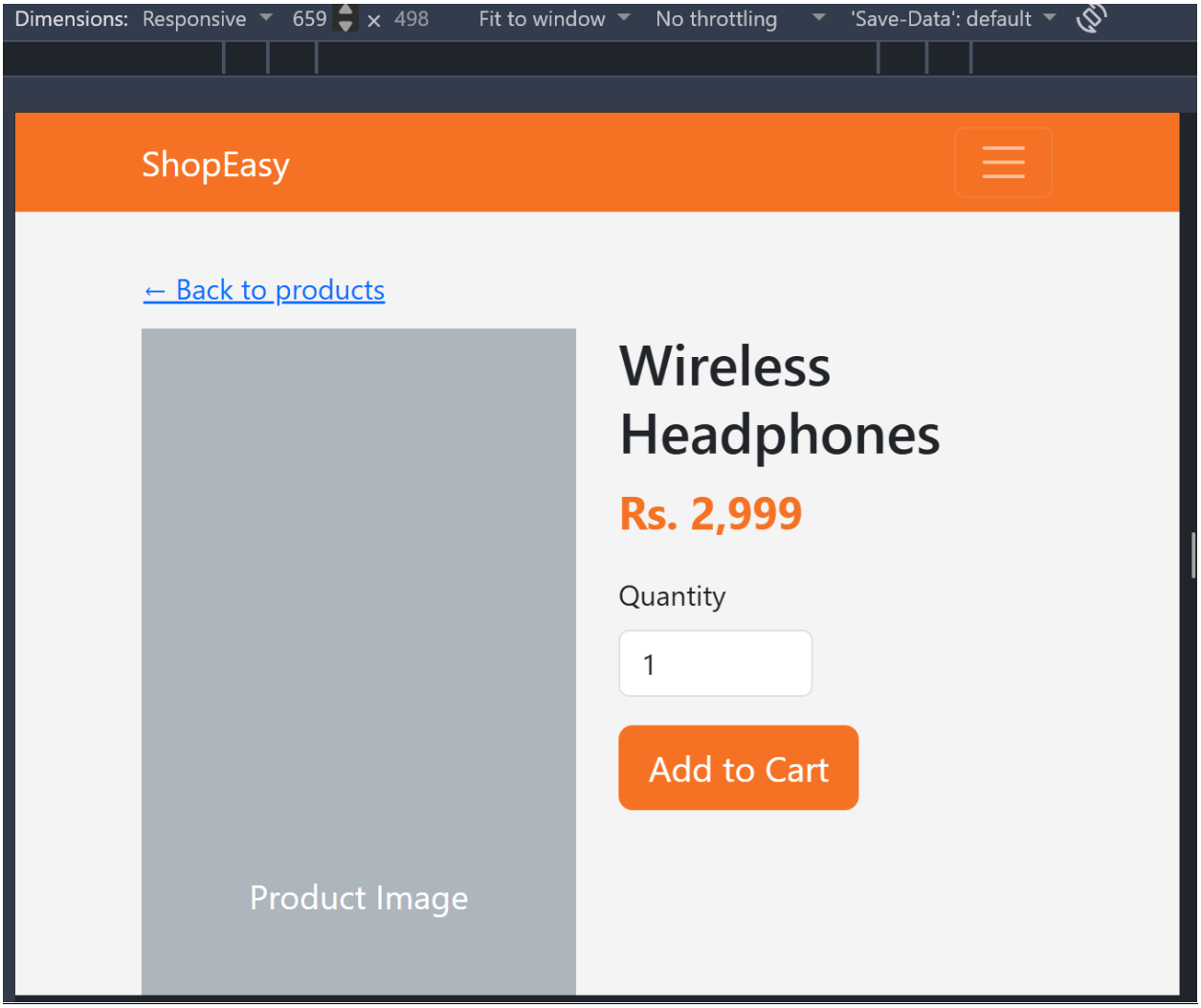

Dimensions: Responsive 659 × 498 Fit to window No throttling 'Save-Data': default
Mobile S – 320px
ShopEasy
Featured Products
All
Audio
Wearables
Footwear
Travel
Home
Image 1
Wireless Headphones
Rs. 2,999

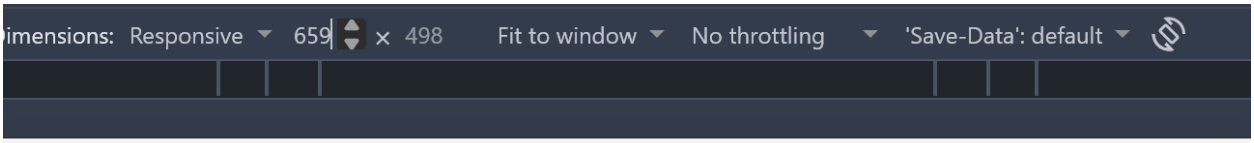

Dimensions: Responsive 1280 x 996 Fit to window No throttling 'Save-Data': default
ShopEasy
Home Products Categories Contact
Cart (0)
Featured Products
All
Audio
Wearables
Footwear
Travel
Home
Image 1
Wireless Headphones
Rs. 2,999
Add to Cart
Image 2
Smart Watch
Rs. 4,999
Add to Cart
Image 3
Backpack
Rs. 1,999
Add to Cart
Image 4
Running Shoes
Rs. 3,499
Add to Cart
Image 5
Sunglasses
Rs. 1,499
Add to Cart
Image 6
Water Bottle
Rs. 899
Add to Cart
Image 7
Desk Lamp
Rs. 1,299
Add to Cart
Image 8
Bluetooth Speaker
Rs. 2,499
Add to Cart


Dimensions: Responsive 1280 x 996 Fit to window No throttling 'Save-Data': default
ShopEasy
Home Products Categories Contact
Cart (0)
Featured Products
All Audio Wearables Footwear Travel Home
Image 1
Image 8
Wireless Headphones
Rs. 2,999
Add to Cart
Bluetooth Speaker
Rs. 2,499
Add to Cart

Task 2:


mensions: Responsive 375 × 498 Fit to window No throttling 'Save-Data': default
Mobile S – 320px
ShopEasy
← Back to products
Product Image


Dimensions: Responsive 659 × 498 Fit to window No throttling 'Save-Data': default
ShopEasy
← Back to products
Product Image
Wireless Headphones
Rs. 2,999
Quantity
1
Add to Cart



Description

Comfortable over-ear wireless headphones with clear sound, a long-lasting battery, and a built-in microphone for calls. Great for music, study, and travel.

Wireless Headphones

Rs. 2,999

The soft memory-foam ear cushions stay comfortable through long listening sessions, and the adjustable steel-reinforced headband fits most head sizes without pinching.

Bluetooth 5.3 gives a stable connection up to 10 metres, and the headphones pair with two devices at once, so you can switch from your laptop to your phone without reconne

Quantity

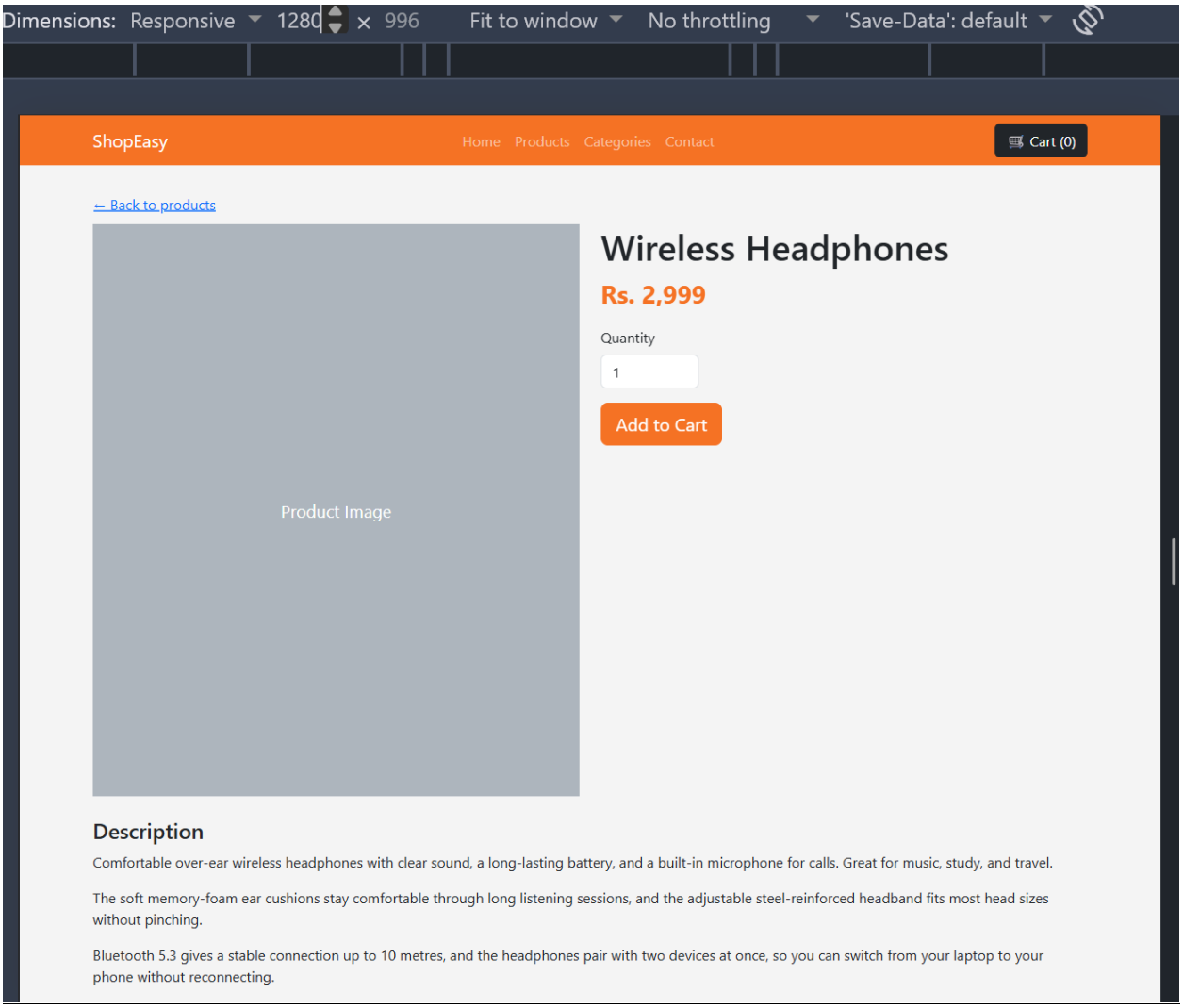
1

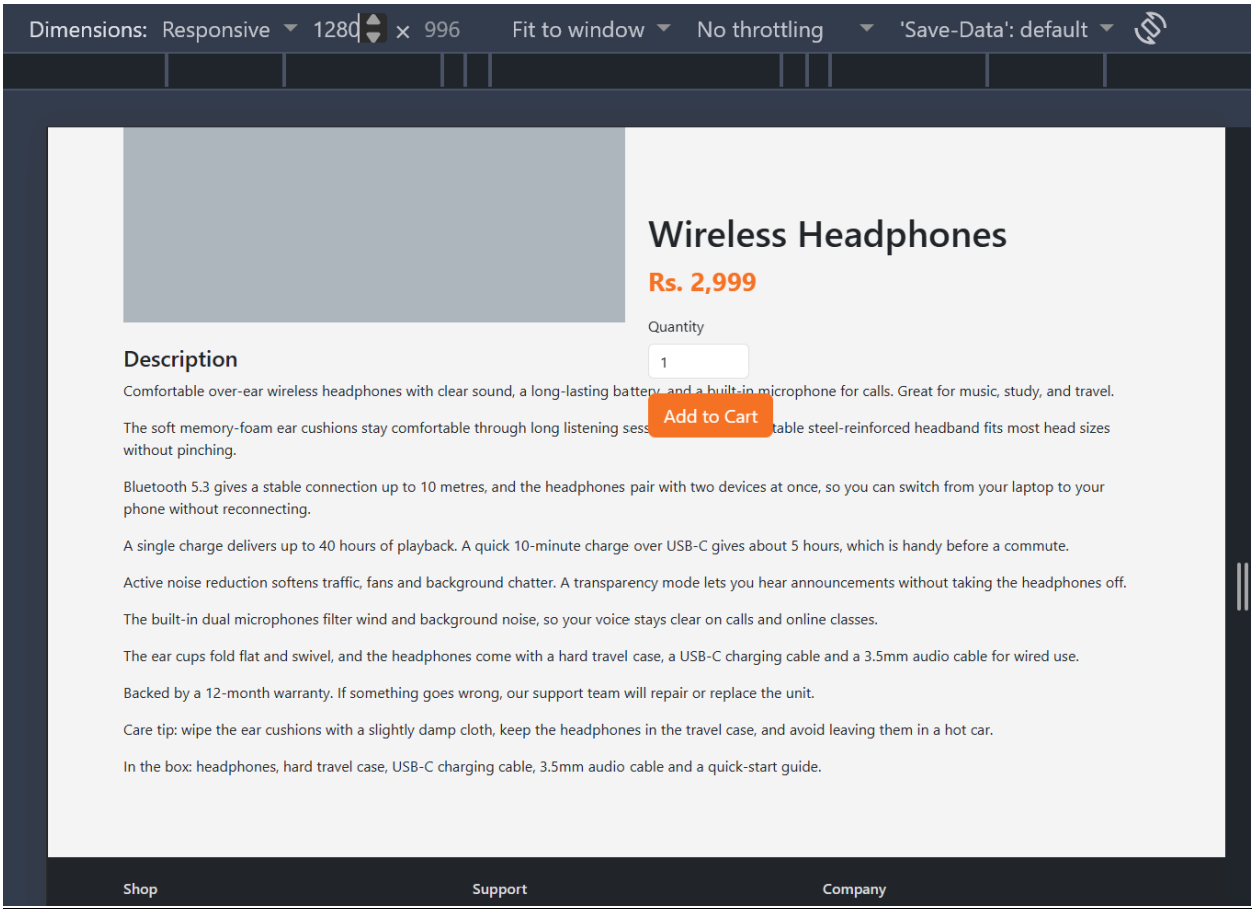
A single charge delivers up to 40 hours of playback. A quick 10-minute charge over USB-C gives about 5 hour efore a commute.

Add to Cart

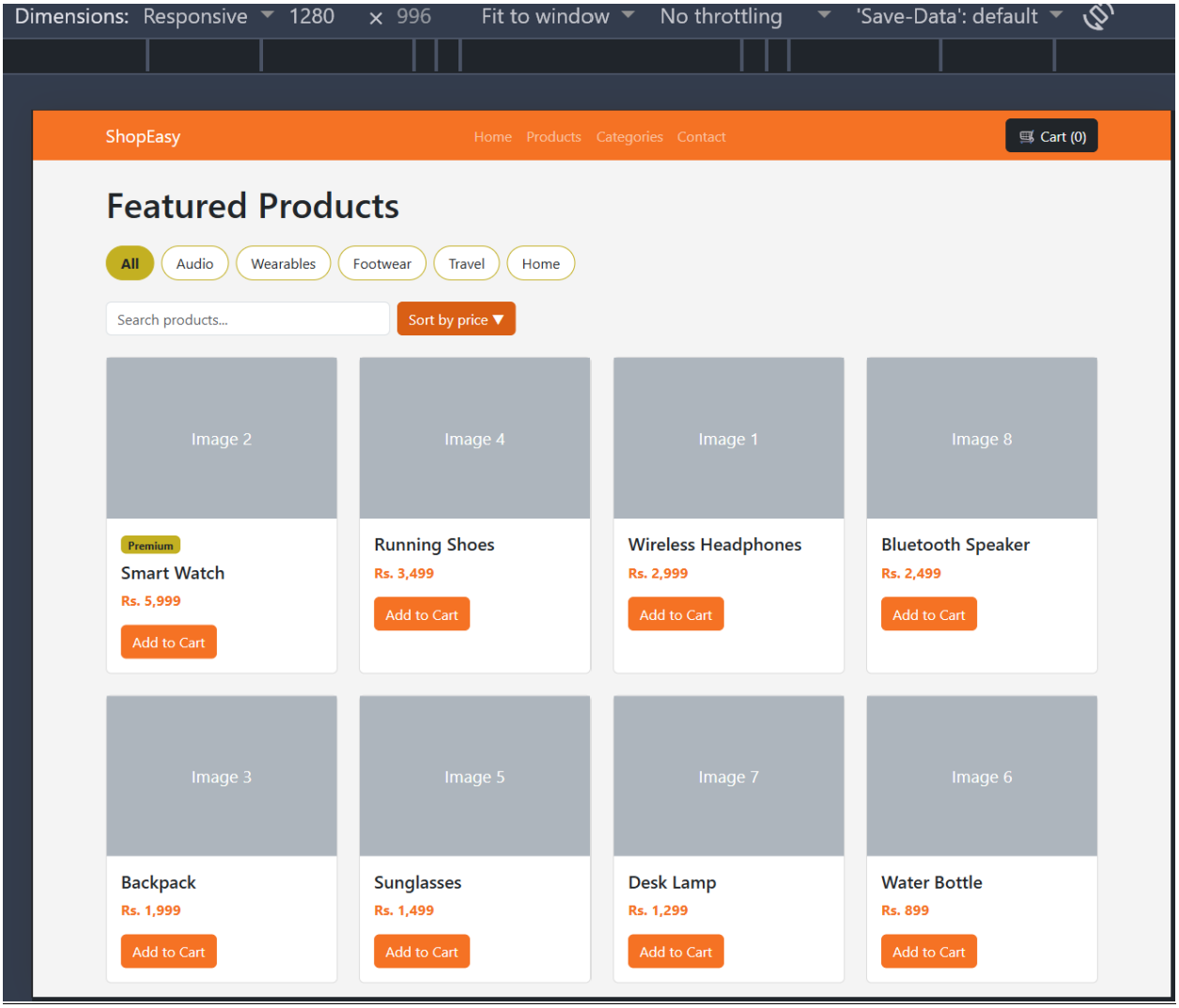
Active noise reduction softens traffic, fans and background chatter. A transparency mode lets you hear announcements without taking the headphones off.

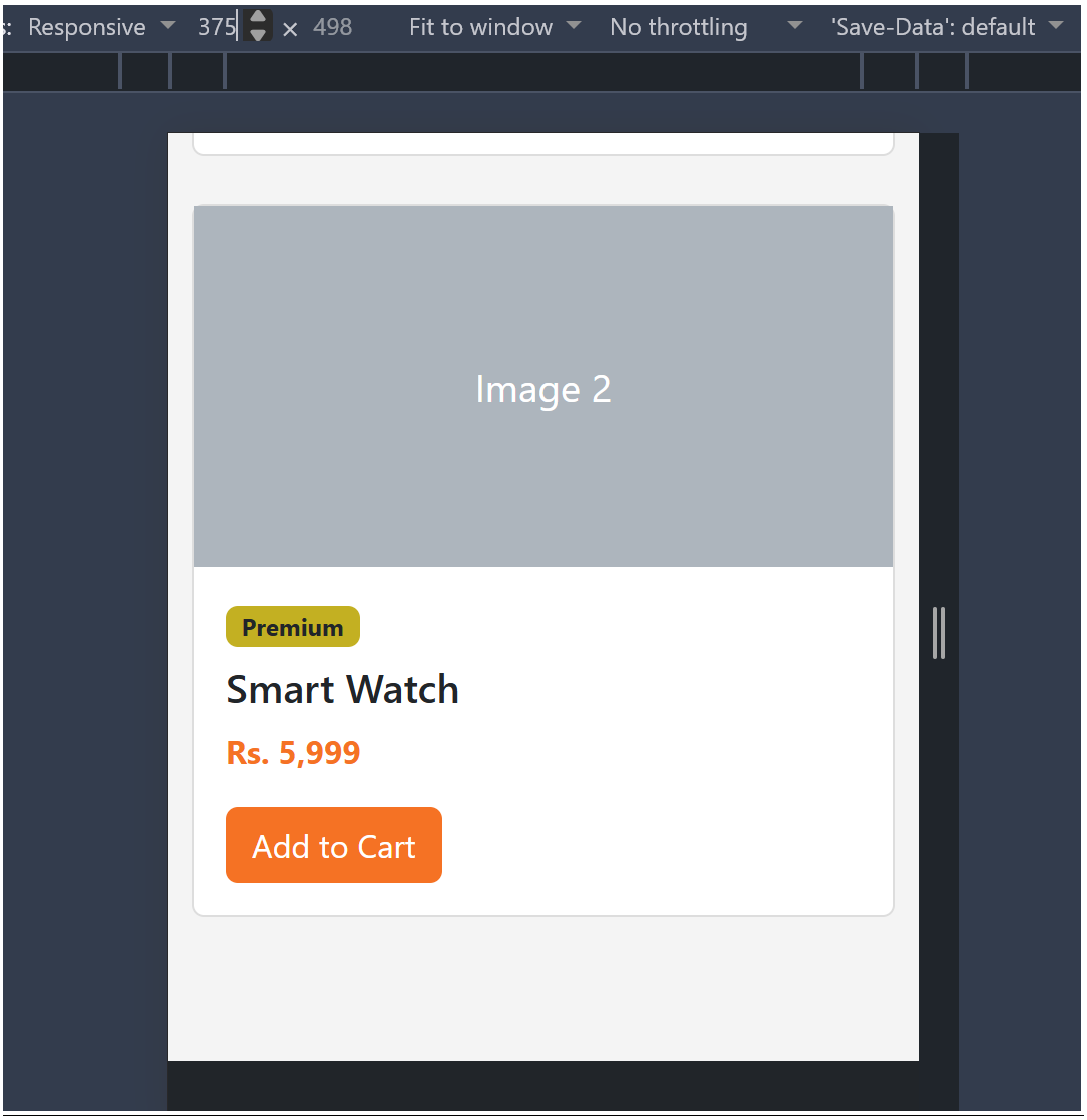
The built-in dual microphones filter wind and background noise, so your

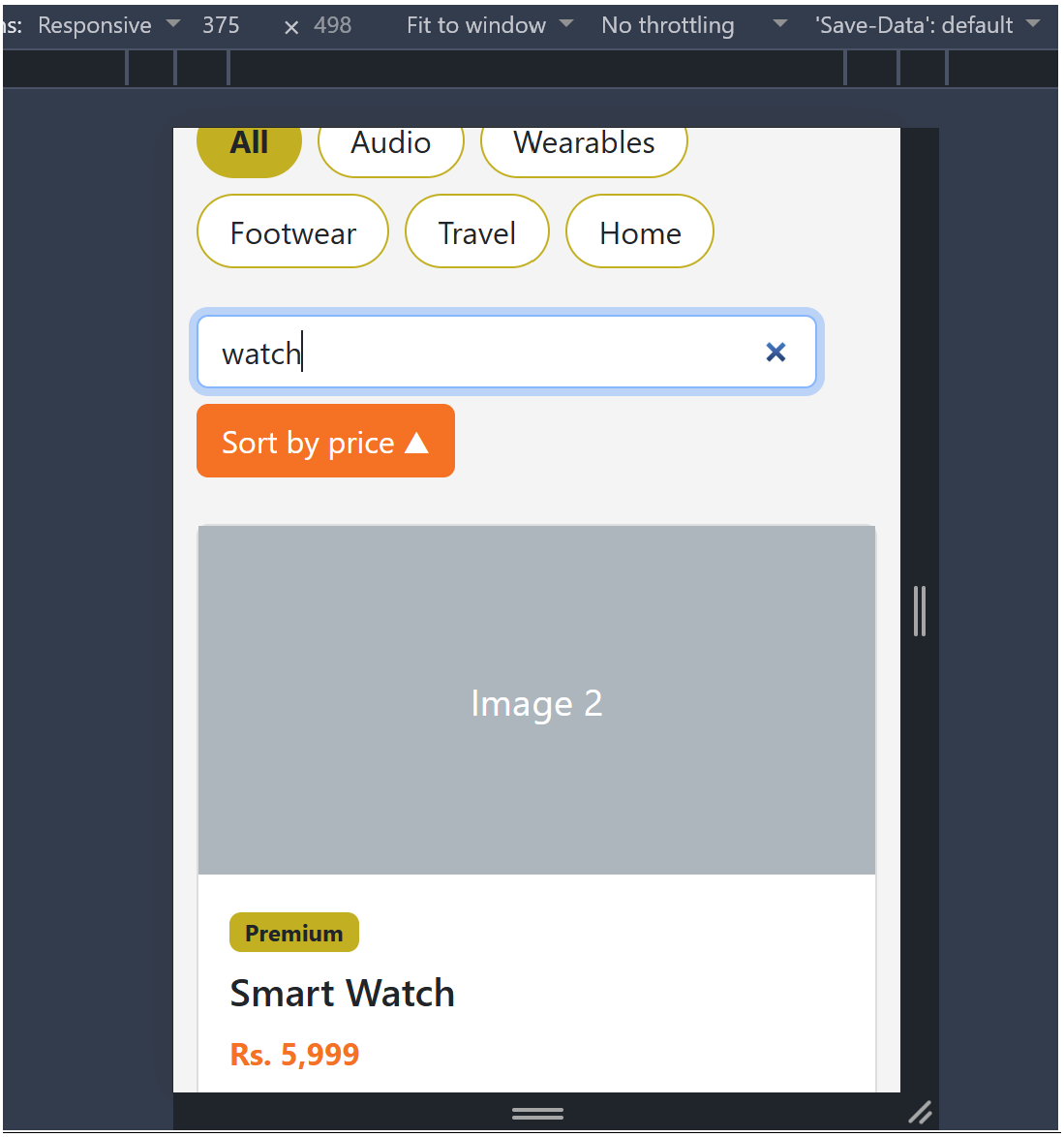

Dimensions: Responsive 1280 x 996 Fit to window No throttling 'Save-Data': default
ShopEasy
Home Products Categories Contact
Cart (0)
← Back to products
Product Image
Wireless Headphones
Rs. 2,999
Quantity
1
Add to Cart
Description
Comfortable over-ear wireless headphones with clear sound, a long-lasting battery, and a built-in microphone for calls. Great for music, study, and travel.
The soft memory-foam ear cushions stay comfortable through long listening sessions, and the adjustable steel-reinforced headband fits most head sizes without pinching.
Bluetooth 5.3 gives a stable connection up to 10 metres, and the headphones pair with two devices at once, so you can switch from your laptop to your phone without reconnecting.

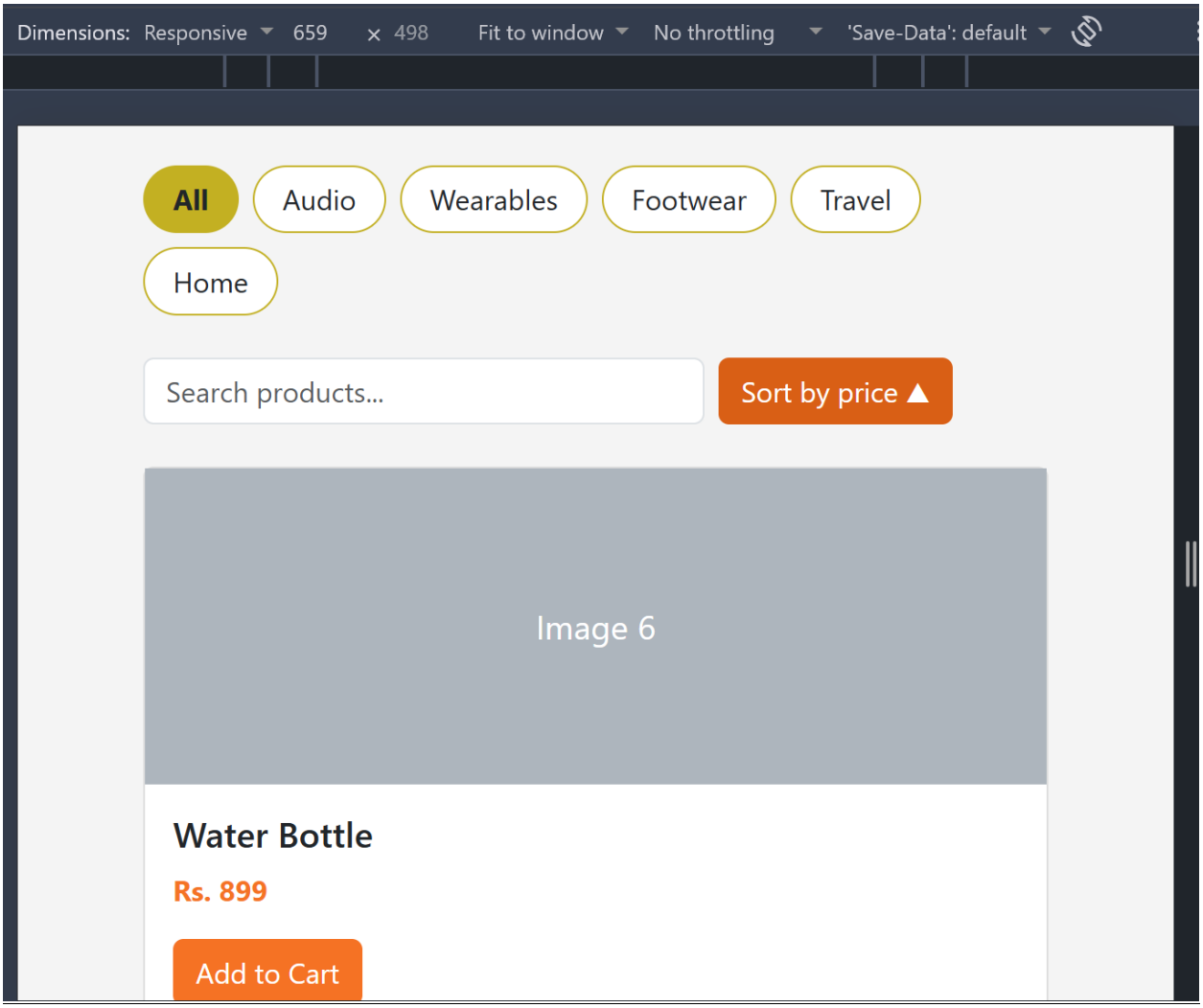

Dimensions: Responsive 1280 × 996 Fit to window No throttling 'Save-Data': default
Wireless Headphones
Rs. 2,999
Quantity
1
Add to Cart
Description
Comfortable over-ear wireless headphones with clear sound, a long-lasting battery, and a built-in microphone for calls. Great for music, study, and travel.
The soft memory-foam ear cushions stay comfortable through long listening sess table steel-reinforced headband fits most head sizes without pinching.
Bluetooth 5.3 gives a stable connection up to 10 metres, and the headphones pair with two devices at once, so you can switch from your laptop to your phone without reconnecting.
A single charge delivers up to 40 hours of playback. A quick 10-minute charge over USB-C gives about 5 hours, which is handy before a commute.
Active noise reduction softens traffic, fans and background chatter. A transparency mode lets you hear announcements without taking the headphones off.
The built-in dual microphones filter wind and background noise, so your voice stays clear on calls and online classes.
The ear cups fold flat and swivel, and the headphones come with a hard travel case, a USB-C charging cable and a 3.5mm audio cable for wired use.
Backed by a 12-month warranty. If something goes wrong, our support team will repair or replace the unit.
Care tip: wipe the ear cushions with a slightly damp cloth, keep the headphones in the travel case, and avoid leaving them in a hot car.
In the box: headphones, hard travel case, USB-C charging cable, 3.5mm audio cable and a quick-start guide.
Shop
Support
Company

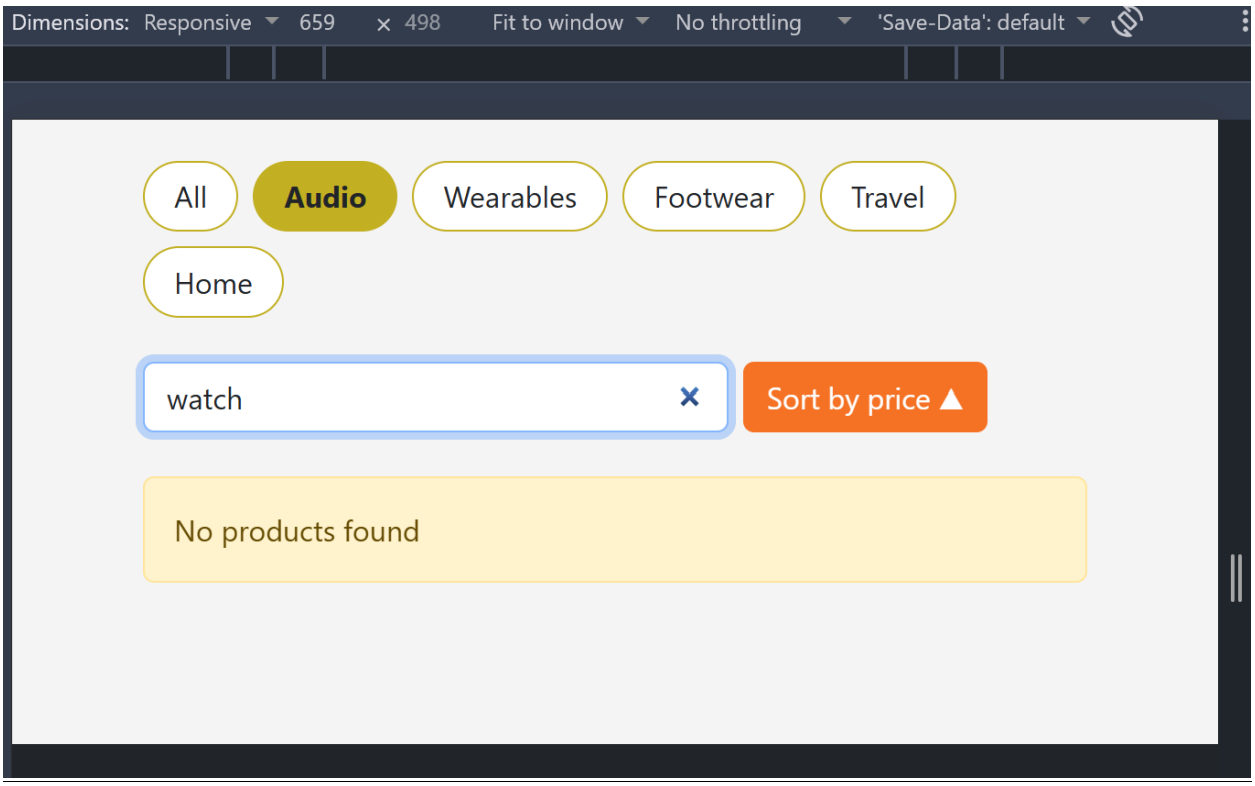
Task 3:

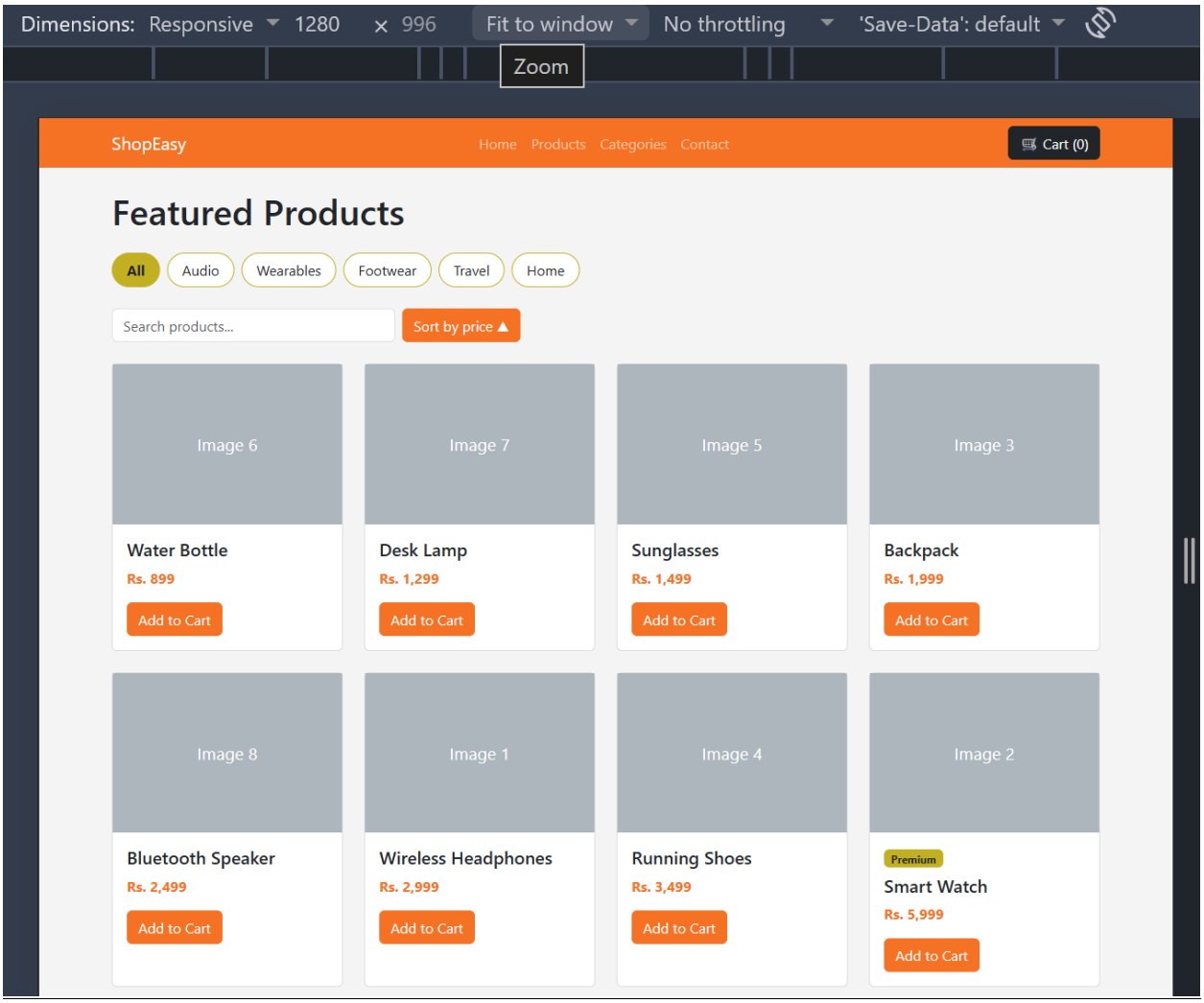

Dimensions: Responsive 1280 × 996 Fit to window No throttling 'Save-Data': default
ShopEasy
Home Products Categories Contact
Cart (0)
Featured Products
All
Audio
Wearables
Footwear
Travel
Home
Search products...
Sort by price ▼
Image 2
Premium
Smart Watch
Rs. 5,999
Add to Cart
Image 4
Running Shoes
Rs. 3,499
Add to Cart
Image 1
Wireless Headphones
Rs. 2,999
Add to Cart
Image 8
Bluetooth Speaker
Rs. 2,499
Add to Cart
Image 3
Backpack
Rs. 1,999
Add to Cart
Image 5
Sunglasses
Rs. 1,499
Add to Cart
Image 7
Desk Lamp
Rs. 1,299
Add to Cart
Image 6
Water Bottle
Rs. 899
Add to Cart


Responsive
375
x
498
Fit to window
No throttling
'Save-Data': default
Image 2
Premium
Smart Watch
Rs. 5,999
Add to Cart


ns: Responsive 375 × 498 Fit to window No throttling 'Save-Data': default
All
Audio
Wearables
Footwear
Travel
Home
watch
Sort by price ▲
Image 2
Premium
Smart Watch
Rs. 5,999


Dimensions: Responsive 659 × 498 Fit to window No throttling 'Save-Data': default
All
Audio
Wearables
Footwear
Travel
Home
Search products...
Sort by price ▲
Image 6
Water Bottle
Rs. 899
Add to Cart


Dimensions: Responsive 659 × 498 Fit to window No throttling 'Save-Data': default
All
Audio
Wearables
Footwear
Travel
Home
watch
Sort by price ▲
No products found


Dimensions: Responsive 1280 × 996 Fit to window No throttling 'Save-Data': default
Zoom
ShopEasy
Home Products Categories Contact
Cart (0)
Featured Products
All
Audio
Wearables
Footwear
Travel
Home
Search products...
Sort by price ▲
Image 6
Water Bottle
Rs. 899
Add to Cart
Image 7
Desk Lamp
Rs. 1,299
Add to Cart
Image 5
Sunglasses
Rs. 1,499
Add to Cart
Image 3
Backpack
Rs. 1,999
Add to Cart
Image 8
Bluetooth Speaker
Rs. 2,499
Add to Cart
Image 1
Wireless Headphones
Rs. 2,999
Add to Cart
Image 4
Running Shoes
Rs. 3,499
Add to Cart
Image 2
Premium
Smart Watch
Rs. 5,999
Add to Cart

Code Added / Changed

Student: Abdullah | Reg. No: FA24-BCS-159 | N = 59, BP = 659px, H = 53, K = 5

Task 1 - CSS-only category filter

Each product wrapper got a data-category (8 cards shown by their opening tags), and the filter radios were added above the grid.

index.html - category on each card (opening tags only)

```
<div class="col" data-category="audio">
<div class="col" data-category="wearables">
<div class="col" data-category="travel">
<div class="col" data-category="footwear">
<div class="col" data-category="wearables">
<div class="col" data-category="travel">
<div class="col" data-category="home">
<div class="col" data-category="audio">
```

index.html - filter radio buttons

```
<div class="filter-bar mb-4" role="radiogroup" aria-label="Filter by category">
      <input type="radio" class="visually-hidden" name="category" id="cat-all" checked>
      <label for="cat-all">All</label>

      <input type="radio" class="visually-hidden" name="category" id="cat-audio">
      <label for="cat-audio">Audio</label>

      <input type="radio" class="visually-hidden" name="category" id="cat-wearables">
      <label for="cat-wearables">Wearables</label>

      <input type="radio" class="visually-hidden" name="category" id="cat-footwear">
      <label for="cat-footwear">Footwear</label>

      <input type="radio" class="visually-hidden" name="category" id="cat-travel">
      <label for="cat-travel">Travel</label>

      <input type="radio" class="visually-hidden" name="category" id="cat-home">
      <label for="cat-home">Home</label>
    </div>
```

styles.css - Task 1

```
/* ---- Task 1: CSS-only category filter ---- */
.filter-bar {
  display: flex;
  flex-wrap: wrap;
  gap: 0.5rem;
}

.filter-bar label {
  padding: 0.4rem 1rem;
  border: 1px solid hsl(53, 70%, 45%);
  border-radius: 2rem;
  background: #fff;
  color: #212529;
  cursor: pointer;
}

/* selected filter uses accent colour H = 53 */
.filter-bar input:checked + label {
  background: hsl(53, 70%, 45%);
  color: #212529;
  font-weight: bold;
}

/* keyboard focus outline */
.filter-bar input:focus-visible + label {
  outline: 2px solid #212529;
  outline-offset: 2px;
```

```
}

/* hide cards that do not match the checked radio.
   display:none on the .col wrapper removes it from the flex row,
   so the remaining cards close up with no empty gaps. */
main:has(#cat-audio:checked)     section .col:not([data-category="audio"]),
main:has(#cat-wearables:checked) section .col:not([data-category="wearables"]),
main:has(#cat-footwear:checked)  section .col:not([data-category="footwear"]),
main:has(#cat-travel:checked)    section .col:not([data-category="travel"]),
main:has(#cat-home:checked)      section .col:not([data-category="home"]) {
  display: none;
}
```

Task 2 - Product page layout with sticky details box

product.html - grid layout markup

```
<!-- Task 2: CSS Grid with named grid-template-areas -->
    <article class="product-layout mt-1">
      <div class="product-media">
        <div class="product-img detail-img">Product Image</div>
      </div>

      <div class="product-details">
        <h1>Wireless Headphones</h1>
        <p class="price fs-3">Rs. 2,999</p>
        <form>
          <label for="qty" class="form-label">Quantity</label>
          <input type="number" id="qty" name="qty" class="form-control qty-input mb-3" value="1"
            min="1" max="10">
          <button type="button" class="btn btn-accent btn-lg">Add to Cart</button>
        </form>
      </div>

      <div class="product-description">
        <h2 class="h4">Description</h2>
        <p>
          Comfortable over-ear wireless headphones with clear sound, a long-lasting battery,
          and a built-in microphone for calls. Great for music, study, and travel.
        </p>
        <p>The soft memory-foam ear cushions stay comfortable through long listening sessions, and
          the adjustable steel-reinforced headband fits most head sizes without pinching.</p>
        <p>Bluetooth 5.3 gives a stable connection up to 10 metres, and the headphones pair with two
          devices at once, so you can switch from your laptop to your phone without
          reconnecting.</p>
        <p>A single charge delivers up to 40 hours of playback. A quick 10-minute charge over USB-C
          gives about 5 hours, which is handy before a commute.</p>
        <p>Active noise reduction softens traffic, fans and background chatter. A transparency mode
          lets you hear announcements without taking the headphones off.</p>
        <p>The built-in dual microphones filter wind and background noise, so your voice stays clear
          on calls and online classes.</p>
        <p>The ear cups fold flat and swivel, and the headphones come with a hard travel case, a
          USB-C charging cable and a 3.5mm audio cable for wired use.</p>
        <p>Backed by a 12-month warranty. If something goes wrong, our support team will repair or
          replace the unit.</p>
        <p>Care tip: wipe the ear cushions with a slightly damp cloth, keep the headphones in the
          travel case, and avoid leaving them in a hot car.</p>
        <p>In the box: headphones, hard travel case, USB-C charging cable, 3.5mm audio cable and a
          quick-start guide.</p>
      </div>
    </article>
```

styles.css - Task 2

```
/* ---- Task 2: product page grid with sticky details box ---- */
.product-layout {
  display: grid;
  gap: 1.5rem;
  grid-template-columns: 1fr;
  grid-template-areas:
    "image"
    "details"
    "description";
}

.product-media       { grid-area: image; }
.product-details     { grid-area: details; }
.product-description { grid-area: description; }

/* BP = 600 + 59 = 659px */
@media (min-width: 659px) {
  .product-layout {
    grid-template-columns: 1fr 1fr;
    grid-template-areas:
```

```
      "image       details"
      "description description";
    align-items: start;   /* lets the details box stay short so sticky can work */
  }

  .product-details {
    position: sticky;
    top: 89px;            /* (59 mod 40) + 70 = 89 */
  }

  .detail-img {
    height: 640px;        /* tall image so the sticky effect is visible while scrolling */
  }
}
```

Task 3 - JavaScript live search and price sort

index.html - search box, sort button, message

```
<!-- Task 3: live search + price sort (handled by script.js) -->
    <div class="d-flex flex-wrap gap-2 mb-4">
      <label for="search" class="visually-hidden">Search products</label>
      <input type="search" id="search" class="form-control search-input" placeholder="Search
        products...">
      <button type="button" id="sort-btn" class="btn btn-accent">Sort by price</button>
    </div>

    <p id="no-results" class="alert alert-warning" hidden>No products found</p>
```

index.html - other small changes

```
<section id="product-grid" class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4">
<p class="price">Rs. 5,999</p>   <!-- Smart Watch: raised above Rs. 5,900 -->
<script src="script.js"></script>   <!-- before </body> -->
```

script.js (new file)

```
// Task 3: live search + price sort + Premium labels
// Works on the product cards already in index.html (no arrays, no innerHTML).

const PREMIUM_LIMIT = 5900;   // N x 100, with N = 59

const grid = document.getElementById('product-grid');
const searchBox = document.getElementById('search');
const sortBtn = document.getElementById('sort-btn');
const noResults = document.getElementById('no-results');
const radios = document.querySelectorAll('input[name="category"]');

let ascending = true;   // first click sorts low -> high

// Read the price from the card itself: "Rs. 2,999" -> 2999
function getPrice(col) {
  const text = col.querySelector('.price').textContent;
  return parseInt(text.replace(/[^0-9]/g, ''), 10);
}

// Product name in lower case, for case-insensitive search
function getName(col) {
  return col.querySelector('.card-title').textContent.toLowerCase();
}

// Category of the checked radio: "all", "audio", "wearables", ...
function getSelectedCategory() {
  const checked = document.querySelector('input[name="category"]:checked');
  return checked.id.replace('cat-', '');
}

// Show a card only if it matches BOTH the search text and the category
function applyFilters() {
  const query = searchBox.value.trim().toLowerCase();
  const category = getSelectedCategory();
  let visibleCount = 0;

  Array.from(grid.children).forEach(function (col) {
    const nameMatches = getName(col).includes(query);
    const categoryMatches = category === 'all' || col.dataset.category === category;
    const show = nameMatches && categoryMatches;

    col.classList.toggle('d-none', !show);
    if (show) {
      visibleCount++;
    }
  });

  // No match: hide the grid and show the message in its place
  grid.hidden = visibleCount === 0;
```

```
  noResults.hidden = visibleCount !== 0;
}

// Live search: runs on every keystroke
searchBox.addEventListener('input', applyFilters);

// Re-check when the category radio changes, so both filters stay in sync
radios.forEach(function (radio) {
  radio.addEventListener('change', applyFilters);
});

// Sort by price: toggles low -> high, then high -> low, and so on
sortBtn.addEventListener('click', function () {
  const cards = Array.from(grid.children);

  cards.sort(function (a, b) {
    return ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a);
  });

  // appendChild moves each existing card to the end, in the sorted order
  cards.forEach(function (card) {
    grid.appendChild(card);
  });

  sortBtn.textContent = ascending ? 'Sort by price \u25B2' : 'Sort by price \u25BC';
  ascending = !ascending;
});

// On page load: add a "Premium" label to products above the limit
Array.from(grid.children).forEach(function (col) {
  if (getPrice(col) > PREMIUM_LIMIT) {
    const label = document.createElement('span');
    label.className = 'badge premium-label mb-2';
    label.textContent = 'Premium';
    col.querySelector('.card-body').prepend(label);
  }
});

applyFilters();
```

styles.css - Task 3

```
/* ---- Task 3: search box and Premium label ---- */
.search-input {
  max-width: 320px;
}

.premium-label {
  background-color: hsl(53, 70%, 45%);
  color: #212529;
}
```